Homework 3 — Study Notes

Toy elliptic-curve encryption, the Caesar cipher, and statistical cryptanalysis

1. Overview

The homework has three parts. (i) Encrypt and decrypt a message with the toy scheme built on the curve of Part 1 (EC-ElGamal), explaining the steps and the role of the keys. (ii) Do the same with the Caesar cipher and compare the two schemes in procedure, key space and statistical behaviour. (iii) Break the Caesar cipher by exploiting the letter distribution of a known language, and test the attack on ciphertexts of different lengths.

2. EC-ElGamal on the toy curve

Setting. Curve E: $y^2 = x^3 + 7 \pmod{17}$, the equation of secp256k1 over a tiny field. E has 17 affine points plus the point at infinity O (18 in total). Base point G = (1, 5), whose order is n = 9, that is nG = O and the multiples kG repeat with period 9.

Group law. For P ≠ Q the slope is $\lambda = (y_Q - y_P)/(x_Q - x_P)$; for P = Q it is $\lambda = 3x_P^2/(2y_P)$. Then $x_R = \lambda^2 - x_P - x_Q$ and $y_R = \lambda(x_P - x_R) - y_P$, all mod 17. The inverse of (x, y) is (x, −y); P + (−P) = O. Scalar multiplication kP is P added k times.

Roles of the keys

Key	Who	Definition	Role
Private key d	Receiver (Bob)	secret integer, 1 ≤ d < n	the only thing that can decrypt
Public key Q	Receiver (Bob)	Q = dG, published	lets anyone encrypt; recovering d from Q is the ECDLP
Ephemeral key r	Sender (Alice)	fresh random integer per block, discarded	randomises the ciphertext; same plaintext, different output

Algorithm

- **Key generation (Bob).** Choose d, compute Q = dG, publish Q.
- **Encoding.** Curve points can hide points, not text. Text → UTF-8 bytes → 4-bit values m ∈ {0,...,15}; each m is mapped to one of 16 fixed curve points M = A[m].
- **Encryption (Alice, knows only Q).** For every block pick r and send the pair $C_1 = rG$, $C_2 = M + rQ$. The term rQ is a mask that hides M.
- **Decryption (Bob, knows d).** Compute $dC_1 = d(rG) = r(dG) = rQ$ — the same mask — then $M = C_2 - dC_1$. Map M back to m and rebuild the bytes.

Why it works. Scalar multiplication commutes: Alice computes r(dG), Bob computes d(rG); both equal (rd)G. An eavesdropper sees G, Q, C_1, C_2 but needs r or d to remove the mask; finding either is a discrete logarithm.

Worked example (one block)

Take d = 4, so Q = 4G = (12, 1). Encrypt the nibble m = 10 (the point M = (8, 14)) with r = 5:

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C1 = rG        = 5·(1, 5) = (12, 16)
rQ             = 5·(12, 1) = (2, 10)
C2 = M + rQ    = (8, 14) + (2, 10) = (15, 4)
dC1            = 4·(12, 16) = (2, 10)        (equals rQ)
M = C2 - dC1   = (15, 4) + (2, 7) = (8, 14)  ->  m = 10
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Limits of the toy. G has only 9 multiples, so Eve can try every $d \in \{1,\ldots,8\}$. On secp256k1 $n \approx 2^{256}$ and brute force is infeasible. Also, textbook ElGamal is malleable and expands the data (two points per 4-bit block); real systems use the curve to agree on a key (ECDH/ECIES) and encrypt with a symmetric cipher.

3. The Caesar cipher

Letters are identified with $\mathbb{Z}_{26}$ (A = 0, …, Z = 25). With secret key k: encryption $c = (m + k) \bmod 26$, decryption $m = (c - k) \bmod 26$. Non-letters are left unchanged. Example, k = 3: HELLO → KHOOR.

Key space: 25 non-trivial keys (≈ 4.7 bits). Exhaustive search over the 25 candidates is instantaneous.

4. Comparison of the two schemes

	Caesar	EC-ElGamal
Type	symmetric: one shared secret k	public-key: Q to encrypt, d to decrypt
Encryption	$m + k$ in $\mathbb{Z}_{26}$	$(rG, M + rQ)$ in the curve group
Decryption	$c - k$	$C_2 - dC_1$
Randomness	deterministic	new r for every block
Ciphertext size	same as the message	two points per 4-bit block
Hard problem	none	discrete logarithm (ECDLP)
Key space	25 keys	toy: 8 keys; secp256k1: $\approx 2^{256}$
Brute force	instant	toy: instant; secp256k1: infeasible

Similarities. Both hide the plaintext by adding an element of a cyclic group ($\mathbb{Z}_{26}$ or the curve group) and remove it by subtracting the same element; both decrypt exactly with the right key and give a well-formed but wrong output with a wrong key.

Statistical properties. Caesar is a relabelling of the alphabet: the number of distinct letters, the entropy, the share of the most frequent symbol and the index of coincidence are identical to those of the plaintext, and word lengths, repeated words and punctuation survive. ElGamal adds a random mask: $C_1 = rG$ does not depend on the message and C_2 is spread over many points, so equal plaintext blocks give different ciphertexts. In the toy the mask takes only 9 values, so the spread is not perfectly uniform.

Symbols (164-letter sample)	Distinct	Entropy (bits)	Top share	Coincidence
Plaintext letters	26	4.21	15.2%	0.062
Caesar ciphertext letters	26	4.21	15.2%	0.062
Plaintext nibbles (328)	16	2.80	37.8%	0.225
ElGamal C_1 points	8	2.99	14.6%	0.125
ElGamal C_2 points	18	4.07	9.1%	0.060

Index of coincidence: probability that two randomly chosen symbols are equal, $\Sigma\, c_i(c_i-1)/(N(N-1))$. Values from the lab page for d = 4, k = 3; the random r makes the ElGamal figures vary slightly between runs.

5. Statistical cryptanalysis of the Caesar cipher

Idea. Caesar rotates the letter histogram without changing its shape. The correct key is the shift that makes the decrypted histogram look like the language's histogram.

Algorithm. Measure letter probabilities p_i on a reference corpus of the known language (English, ≈1400 letters, add-one smoothing so no p_i is 0). For a ciphertext of N letters and each candidate k = 0,...,25:

- decrypt with k and count the letters, giving observed counts O_i;
- compute $\chi^2(k) = \Sigma_i (O_i - N\, p_i)^2 / (N\, p_i)$;
- output the k with the smallest χ^2.

A simpler variant assumes the most frequent ciphertext letter is E and sets k = (letter − E) mod 26; the χ^2 version uses the whole distribution and is therefore much more robust on short texts.

Experiment

Random windows of a different English text (Dickens, Lincoln; not part of the reference corpus) are encrypted with a random key k ∈ {1,...,25} and attacked; 1000 trials per length.

Ciphertext length (letters)	5	10	20	30	50	100	200
Key recovered	≈ 52%	≈ 88%	≈ 98%	≈ 99.8%	100%	100%	100%

Figures from one run; they fluctuate by a few points because windows and keys are random.

Discussion

- **Effectiveness.** Success rises quickly with N because the observed histogram converges to the language distribution; it is already above 95% at 20 letters and essentially certain from 50.
- **Short texts.** With 5–10 letters the sample is too small to show the language profile, and wrong keys may match better (about half of the 5-letter ciphertexts fail). Ties and several plausible candidates are common.
- **Limitations.** The attack needs to know the language and have a reference corpus. It assumes one fixed shift, so it fails on polyalphabetic ciphers such as Vigenère, and on atypical text (lipograms, numbers, a different language).
- **Why it works at all.** The cipher preserves the distribution up to a rotation. A scheme that hides the distribution, like the randomised ElGamal blocks, cannot be attacked this way.
- **Caveat on the experiment.** The test text is short (≈ 760 letters), so long windows overlap heavily and the high-length results are less independent than they appear.

6. Summary

Caesar is a keyed substitution with 25 keys and no statistical protection, so a few dozen letters of ciphertext are enough to recover the key. EC-ElGamal needs two related keys and a hard problem, randomises every block and hides the plaintext statistics; the toy version is as breakable as Caesar only because its group has 9 elements, whereas on secp256k1 the key space is beyond brute force.